

main.c

Run

Output

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 int main()
4 {
5     int request[50];
6     int n, head, i;
7     int totalMovement = 0;
8     float averageMovement;
9     printf("Enter number of disk requests: ");
10    scanf("%d", &n);
11    printf("Enter the disk request queue:\n");
12    for(i = 0; i < n; i++)
13    {
14        scanf("%d", &request[i]);
15    }
16    printf("Enter initial head position: ");
17    scanf("%d", &head);
18    printf("\nSeek Sequence:\n");
19    printf("%d", head);
20    for(i = 0; i < n; i++)
21    {
22        totalMovement += abs(head - request[i]);
23        head = request[i];
24        printf(" -> %d", head);
25    }
26    averageMovement = (float)totalMovement / n;
27    printf("\n\nTotal Head Movement = %d cylinders\n", totalMovement);
28    printf("Average Head Movement = %.2f cylinders\n", averageMovement);
29    return 0;
30 }
```

Enter number of disk requests: 7
Enter the disk request queue:
82 170 43 140 24 16 190
Enter initial head position: 50

Seek Sequence:
50 -> 82 -> 170 -> 43 -> 140 -> 24 -> 16 -> 190

Total Head Movement = 642 cylinders
Average Head Movement = 91.71 cylinders

=== Code Execution Successful ===